

Practice Quiz 07B — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 07

State the epsilon-delta definition and use it once.

1. (a) State, with all quantifiers, the definition of $\lim_{x \rightarrow c} f(x) = L$.

Solution

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x (0 < |x - c| < \delta \rightarrow |f(x) - L| < \varepsilon). \quad \blacksquare$$

- (b) For $f(x) = -2x$, prove from the definition that $\lim_{x \rightarrow 3} f(x) = -6$.

Solution Let $\varepsilon > 0$ and choose $\delta = \varepsilon/2 > 0$. Let x be arbitrary and suppose $0 < |x - 3| < \delta$. Then

$$|f(x) - (-6)| = |-2x + 6| = 2|x - 3| < 2\delta = \varepsilon.$$

This verifies the definition, so $\lim_{x \rightarrow 3} (-2x) = -6$. $\blacksquare$